Lecture 11

THIRD SCENARIO

3) $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, σ_1^2, σ_2^2 UNKNOWN , $X \perp\!\!\!\perp Y$ INDEPENDENT

$\underline{X} = (X_1, \dots, X_{m_1})$ IID FROM POP. 1

$\underline{Y} = (Y_1, Y_2, \dots, Y_{m_2})$ IID FROM POP. 2

$$T = \frac{(\bar{X}-\bar{Y}) - (\mu_1-\mu_2)}{\sqrt{\frac{S_1^2}{m_1} + \frac{S_2^2}{m_2}}} \sim t_v$$

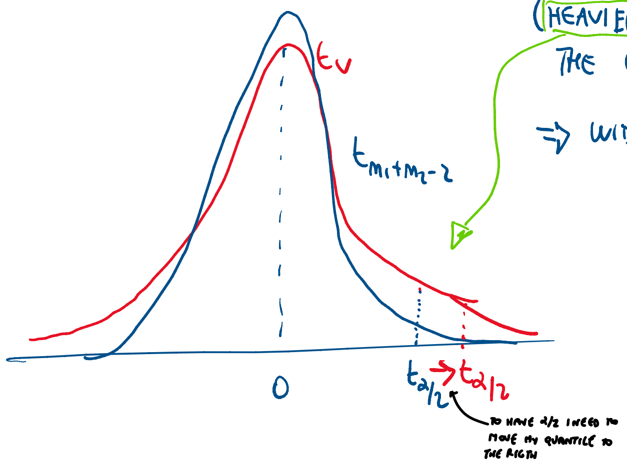
t-STUDENT WITH v DEGREES OF FREEDOM

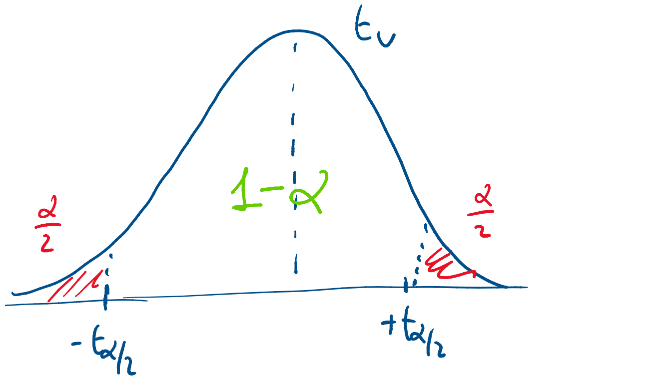
DF < m_1+m_2-2

(HEAVIER TAILS THAN IN THE CASE WITH $\sigma_1^2=\sigma_2^2$ ASSUMED)

$$\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{S_c\sqrt{\frac{1}{m_1}+\frac{1}{m_2}}} \sim t_{m_1+m_2-2}$$

$\Rightarrow$ WIDER CI (CONFIDENCE INTERVAL) AT A GIVEN $1-\alpha$





$$P(-t_{\alpha/2} < T < +t_{\alpha/2}) = 1-\alpha$$

$$P\left(-t_{\alpha/2} < \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{S_1^2/m_1 + S_2^2/m_2}} < +t_{\alpha/2}\right) = 1-\alpha$$

... Skip STEPS TO HAVE $P(L_1 < \underset{\mu_1-\mu_2}{\theta} < L_2)$ ...

$$\boxed{CI_{1-\alpha}(\mu_1-\mu_2) = \left[(\bar{X}-\bar{Y}) \mp t_{\alpha/2}\sqrt{S_1^2/m_1 + S_2^2/m_2}\right]}$$

LENGTH: $L_2 - L_1 = 2\, t_{\alpha/2}\sqrt{S_1^2/m_1 + S_2^2/m_2}$ RANDOM

EX: X: PRICE OF 1 kg BREAD IN MILAN, $m_1 = 25$ OBSERVATIONS

Y: PRICE OF 1 kg BREAD IN MODENA, $m_2 = 22$ OBSERVATIONS

SUMMARY STATS: $\bar{x} = 3.5$ $\bar{y} = 3.8$ (SAMPLE MEAN) $\sum_i (x_i-\bar{x})^2 = 2.4$ $\sum_i (y_i-\bar{y})^2 = 1.35$

FIND THE TRUE MEAN PRICE OF BREAD IN MILAN MINUS THE TRUE MEAN PRICE OF BREAD IN MODENA

FIND $CI_{95\%}(\mu_1-\mu_2)$ ASSUMING $\underline{\sigma_1^2 = \sigma_2^2}$ ASSUMPTION.

$95\% = 1-\alpha$

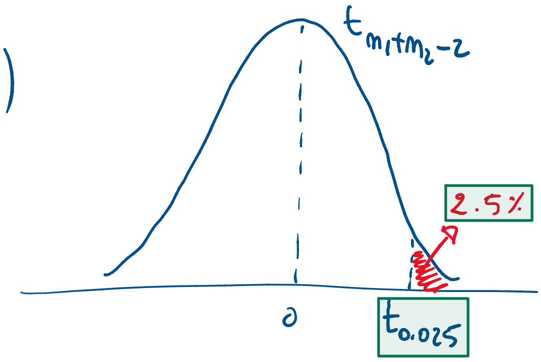
$\Rightarrow$ CASE 2) ABOVE (LAST LECTURE) $\Rightarrow$ UNIQUE ESTIMATOR S_c^2

$$S_c^2 = \frac{(m_1-1)S_1^2 + (m_2-1)S_2^2}{m_1+m_2-2} = \frac{\cancel{(m_1-1)}\cancel{\frac{1}{m_1-1}}\sum_i (x_i-\bar{x})^2 + \cancel{(m_2-1)}\cdot\cancel{\frac{1}{m_2-1}}\sum_i (y_i-\bar{y})^2}{m_1+m_2-2}$$

$$= \frac{\sum_i (x_i-\bar{x})^2 + \sum_i (y_i-\bar{y})^2}{m_1+m_2-2} = \frac{2.4 + 1.39}{25+22-2} = 0.09 \Rightarrow S_c = \sqrt{S_c^2} = 0.3$$

BECAUSE CI IS 95% ⇒ $1-\alpha = 95 \Rightarrow \alpha = 0.05$

$$t_{\alpha/2} = t_{0.05/2} = \boxed{t_{0.025}} = qt(\underbrace{1-0.025}_{\text{THE AREA ON THE LEFT}}, \underbrace{25+22-2}_{\text{DEGREES OF FREEDOM}}) = 2.0141$$



$$CI_{95\%}(\mu_1-\mu_2) = \left[(\bar{x}-\bar{y}) \mp t_{0.025}\cdot S_c\sqrt{\frac{1}{m_1}+\frac{1}{m_2}}\right]$$

⇒ CI IS THE SET OF REASONABLE VALUES ASSIGNED TO THE PARAMETER

$$= \left[(3.5-3.8) \mp 2.0141\cdot 0.3\sqrt{\frac{1}{25}+\frac{1}{22}}\right]$$

$$= [-0.477, -0.123]$$

AT THIS LEVEL OF CONFIDENCE, BREAD PRICES ARE HIGHER IN MODENA (BECAUSE ARE NEGATIVE)

IF $m_1 \uparrow$ OR $m_2 \uparrow$

⇒ MoE ↓ ⇒ STRICTER CI (CENTER DOES NOT CHANGE)

MARGIN OF ERROR

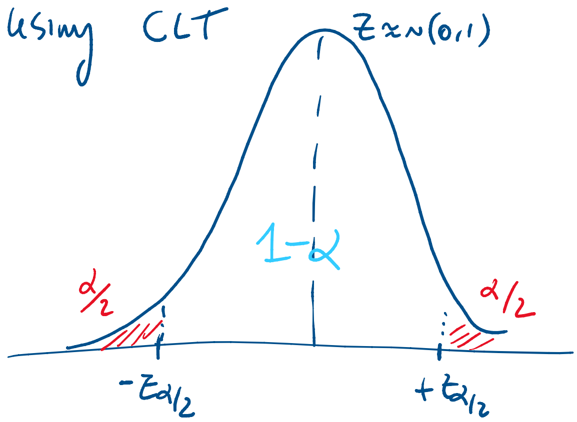
⇒ STILL WHOLE CI ON NEGATIVES

$CI_{1-\alpha}(\mu_1-\mu_2)$, $X \sim$ UNKNOWN, $Y \sim$ UNKNOWN, LARGE SAMPLES

CENTRAL LIMIT THEOREM

PQ: $Z = \dfrac{(\bar{x}-\bar{y})-(\mu_1-\mu_2)}{\sqrt{S_1^2/m_1 + S_2^2/m_2}} \approx N(0,1)$ using CLT

PIVOTAL QUANTITY



$$P(-z_{\alpha/2} < Z < +z_{\alpha/2}) = 1-\alpha$$

... USUAL STEPS ...

$$CI_{1-\alpha}(\mu_1-\mu_2) \approx \left[(\bar{x}-\bar{y}) \mp z_{\alpha/2}\sqrt{S_1^2/m_1 + S_2^2/m_2}\right]$$

BECAUSE OF LARGE SAMPLE WE CAN APPLY THE CENTRAL LIMIT THEOREM AND SO WE PRETEND OUR DATA ARE NORMAL

IMPORTANT SPECIAL CASE: LARGE SAMPLES WITH BERNOULLI POPULATIONS

$$X \sim Bern(p_1) \qquad\qquad Y \sim Bern(p_2)$$

$$V(X) = p_1(1-p_1) \qquad\qquad V(Y) = p_2(1-p_2)$$

$$V(\bar{X}) = \frac{p_1(1-p_1)}{m_1} \qquad\qquad V(\bar{Y}) = p_2(1-p_2)/m_2$$

$$\widehat{V(\bar{X})} = \frac{\bar{X}(1-\bar{X})}{m_1} \qquad\qquad \widehat{V(\bar{Y})} = \bar{Y}(1-\bar{Y})/m_2$$

$$\widehat{V(\bar{X}-\bar{Y})} = \widehat{V(\bar{X})} + \widehat{V(\bar{Y})} = \frac{\bar{X}(1-\bar{X})}{m_1} + \frac{\bar{Y}(1-\bar{Y})}{m_2}$$

PIVOTAL QUANTITY

$$\Rightarrow PQ = \frac{(\bar{X}-\bar{Y})-(p_1-p_2)}{\sqrt{\frac{\bar{X}(1-\bar{X})}{m_1}+\frac{\bar{Y}(1-\bar{Y})}{m_2}}} \approx N(0,1) \quad \text{USING CLT}$$

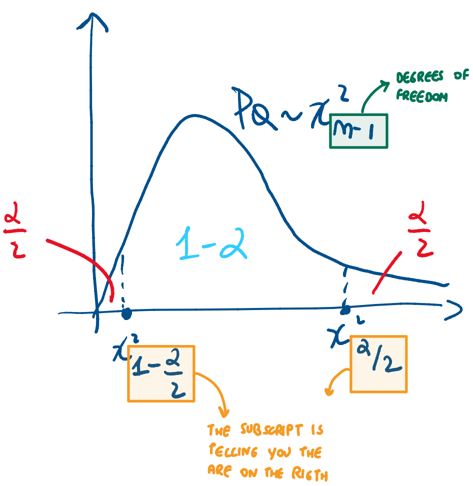
$$\ldots \quad \boxed{CI_{1-\alpha}(p_1-p_2) \approx \left[(\bar{X}-\bar{Y}) \mp z_{\alpha/2}\sqrt{\frac{\bar{X}(1-\bar{X})}{m_1}+\frac{\bar{Y}(1-\bar{Y})}{m_2}}\right]}$$

$$\boxed{CI_{1-\alpha}(\sigma^2), \; X \sim N(\mu,\sigma^2), \; \mu \text{ UNKNOWN}}$$

$$PQ = \frac{(m-1)S^2}{\sigma^2} \sim \boxed{\chi^2_{m-1} \quad \text{CHI-SQUARE DISTRIBUTED RV}}$$

$$= \frac{\cancel{(m-1)} \cdot \frac{1}{\cancel{(m-1)}} \sum_i (X_i-\bar{X})^2}{\sigma^2} = \frac{\sum_i (X_i-\bar{X})^2}{\sigma^2} = \sum_i \left(\frac{X_i-\bar{X}}{\sigma}\right)^2$$

CHI-SQUARED DISTRIBUTION
• CAN ONLY TAKE POSITIVE VALUES



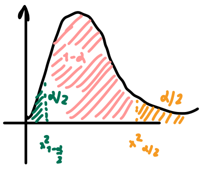
TWO QUANTILES

$$\chi^2_{\frac{\alpha}{2}} = qchisq\left(1-\frac{\alpha}{2}, m-1\right)$$

$$\chi^2_{1-\frac{\alpha}{2}} = qchisq\left(\alpha/2, m-1\right)$$

$$P\left(\boxed{\chi^2_{1-\alpha/2}} < PQ < \boxed{\chi^2_{\alpha/2}}\right) = 1-\alpha$$

$\Leftarrow$ THE PROBABILITY OF THE PIVOTAL QUANTITY TO BE BETWEEN THE LOWER CUTOFF $\chi^2_{1-\alpha}$ AND THE UPPER CUTOFF $\chi^2_{\alpha/2}$ IS $1-\alpha$



$$P\left(\chi^2_{1-\frac{\alpha}{2}} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\alpha/2}\right) = 1-\alpha$$

$$P\left(\frac{\chi^2_{1-\alpha/2}}{(n-1)S^2} < \frac{1}{\sigma^2} < \frac{\chi^2_{\alpha/2}}{(n-1)S^2}\right) = 1-\alpha$$

$$P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}}}\right) = 1-\alpha$$

L_1 LOWER BOUND — θ PARAMETER — L_2 UPPER BOUND

$$\boxed{CI_{1-\alpha}(\sigma^2) = \left[\frac{(n-1)S^2}{\chi^2_{\alpha/2}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right]}$$

⚠ COMMON MISTAKE: INVERT THESE

WE LOST THE STRUCTURE OF POINT ESTIMATOR $\mp$ MARGIN OF ERROR

IF ON THE OTHER HAND: WE ASSUME μ IS KNOWN, (WE KNOW THE POPULATION MEAN)

INSTEAD OF S^2, WE USE $\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\mu)^2$

SAME AS BEFORE BUT REPLACE S^2

BEFORE $PQ = \frac{(n-1)S^2}{\sigma^2}$

$$\Rightarrow PQ = \frac{\cancel{(n-1)} \cdot \boxed{\frac{1}{\cancel{n-1}}\sum_i (X_i-\mu)^2}}{\sigma^2} = \sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 \sim \chi^2_n$$

CHI SQUARED

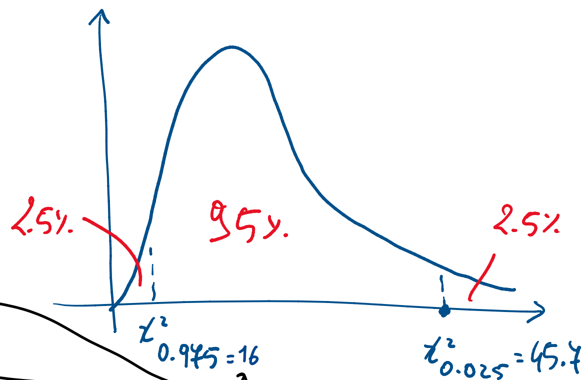
⚠ WE INCREASED THE DEGREE OF FREEDOM, BEFORE IT WAS χ^2_{n-1}

$\Rightarrow CI_{1-\alpha}(\sigma^2)$ AS BEFORE, BUT WITH QUANTILES FROM χ^2_n

EX: $n=30$, $X \sim N(\mu, \sigma^2)$. $\sum_i (X_i-\bar{X})^2 = 90$. μ UNKNOWN

$\boxed{\text{FIND } CI_{95\%}(\sigma^2)}$.

$n = 30 \rightarrow \chi^2_{29}$ IS DISTRIB. OF PQ



$\chi^2_{0.025} = \text{qchisq}(1-0.025, 29) = 45.7$

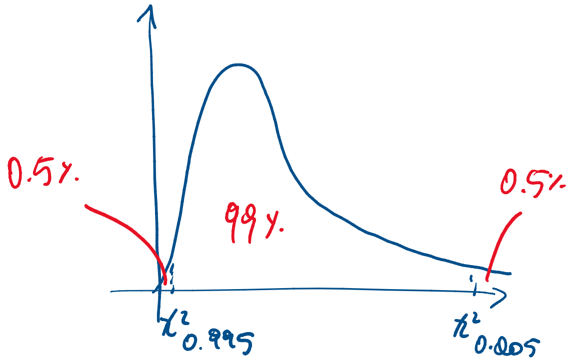
$\chi^2_{0.975} = \text{qchisq}(1-0.975, 29) = 16$

$(n-1)\frac{1}{n-1}\sum_i (X_i-\bar{X})^2$

$$CI_{95\%}(\sigma^2) = \left[\frac{(n-1)S^2}{\chi^2_{0.025}}, \frac{\boxed{(n-1)S^2}}{\chi^2_{0.975}}\right] = \left[\frac{90}{45.7}, \frac{90}{16}\right] = [1.97, 5.6]$$

IF YOU WANT TO INCREASE THE CONFIDENCE THAT YOUR INTERVAL INCLUDE THE TRUE PARAMETER YOU NEED TO INCREASE THE INTERVAL $\Rightarrow$ BIGGER INTERVAL MORE SECURE

$CI_{99\%}(\sigma^2)$ WILL BE WIDER:



$\chi^2_{0.025}$

$$\chi^2_{0.005} = qchisq(1-0.005, 29) = 52.34 > 45.7$$

$\chi^2_{0.975}$

$$\chi^2_{0.995} = qchisq(1-0.995, 29) = 13.12 < 16$$

$$\Rightarrow CI_{99\%}(\sigma^2) = \left[\frac{90}{52.34}, \frac{90}{13.12}\right] = [1.7, 6.9]$$

NOW WIDER

BEFORE:

$CI_{95\%}(\sigma^2) = [1.97, 5.6]$